

National University of Computer and Emerging Sciences, Lahore Campus



Course: Applied Thermodynamics
 Program: BS(Electrical Engineering)
 Duration: 60 Minutes
 Paper Date: 21-Sep-24
 Section: ALL
 Exam: Mid1 Examination

Course Code: ME2002
 Semester: F 2024
 Total Marks: 40
 Weight: 15%
 Page(s): 4
 Questions: 2

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Instruction/Notes:

- > Use valid assumptions if needed, don't ask any questions.
- > Your answers should be correct up to three decimal places with appropriate units.
- > Values read from tables should have the same number of significant figures as mentioned in the tables.
- > This is an open book, and open notes exam

CLO 1: Analyze a thermodynamics process

Q1 A closed system contains 5 kg of water in a Boiler. Initially, the water has a temperature of 25°C and a specific volume of $1.25616 \text{ m}^3/\text{kg}$. The system operates in two stages: [20]

- Stage 1: The system is heated until the pressure inside the boiler reaches 200 kPa keeping volume constant.
- Stage 2: The system is then cooled from a temperature of 150°C , allowing the pressure and temperature to change, but keeping volume constant.

- a) Describe the phase and conditions of the water at $v = 1.25616 \text{ m}^3/\text{kg}$ and $T = 25^{\circ}\text{C}$?
- b) When the pressure inside the boiler reaches 200 kPa, what will be the state of the water? Determine the corresponding temperature.
- c) After the system cools from 150°C , what will be the total volume V of the water in the system? Describe the state of the water at this point.
- d) If the water is a saturated mixture of vapor and liquid, what is its quality at $T = 100^{\circ}\text{C}$? Also, illustrate this state and quality on a $T-v$ diagram.

CLO 1: Analyze a thermodynamics process

Q2 Urea plants use ammonia as a working substance. In such a set up Ammonia is at 10°C with a mass of 300 grams in a piston cylinder with an initial volume of 0.03 m^3 . The piston has a design such that a pressure of 900 kPa will float it. Now the ammonia is slowly heated to 50°C . Identify the states involved, find the final pressure and volume and draw this entire process on a Pv diagram. [20]